BTS-IV-04.19-0376 Reg. No. | | | | | | | | |

B

B.Tech. Degree IV Semester Examination April 2019

CS 1406 DATA COMMUNICATIONS

(2012 Scheme)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer ***ALL*** questions)

(8 × 5 = 40)

I. (a) List out the advantages of digital transmission over analog transmission.
(b) Define channel capacity. A data is to be transmitted over PSTN using 8 levels per signal elements. If the bandwidth is 3000 HZ, deduce maximum data transfer rate.
(c) Illustrate PCM technique with the help of a neat diagram.
(d) Exemplify any two block coding schemes used for digital signal transmission.
(e) Summarize the three different transmission modes possible in HDLC.
(f) Analyse the significance of Hamming Code in Forward Error correction using a suitable example.
(g) Write short note on CDMA.
(h) Discuss the major components of a telephone network.

PART B

(4 × 15 = 60)

II. (a) Summarize the various transmission impairments. (10)
(b) Explain wireless transmission. (5)

OR

III. (a) Describe the components in communication model with a neat diagram. (5)
(b) Explain any three different types of guided transmission media. (10)

IV. (a) Define line coding. Encode the data 110110 using NRZ-L, NRZ-I, Manchester, Differential Manchester and AMI. (10)
(b) Differentiate AM and FM. (5)

OR

V. (a) Construct a Huffman code tree with probabilities given as 0.4, 0.19, 0.16, 0.15, 0.1. Also find the code for each symbol and calculate its average code size and variance. (9)
(b) Discuss any two variants of PSK technique. (6)

VI. (a) A series of 8 bit message 11100110 is transmitted across a data link using a CRC for error detection. A generator polynomial of 11001 is to be used. Illustrate the following: (10)
(i) CRC Generation Process
(ii) CRC Checking Process
(b) Explain the concept of redundancy in error detection. (5)

OR

VII. Summarize the three categories of ARQ error control mechanism. (15)

VIII. (a) Explain the role of CM and CMTS in cable networks. (5)
(b) With the help of necessary figures explain transmitter and receiver system in frequency hopping spread spectrum. (10)

OR

IX. (a) Compare frequency division multiplexing and time division multiplexing. (5)
(b) Explain FDM with the help of necessary sketches. (10)

***